

**Rensselaer Polytechnic Institute**  
**Department of Electrical, Computer, and Systems Engineering**  
**ECSE 2500: Engineering Probability, Spring 2026**  
**Homework #1 Solutions**

1. (a) Will could visit the Upside Down any number of times; the sample space is  $S = \{0, 1, \dots, \infty\}$ .
- (b) Similarly, Eleven and Max could visit Vecna's mind any number of times, so the difference between these numbers could be arbitrarily big or small; the sample space is  $S = \{0, \pm 1, \pm 2, \dots, \pm \infty\}$ .
- (c) We have to allocate the kids between the two possible minds; there are 6 total outcomes in the sample space:  $\{(5, 0), (4, 1), (3, 2), (2, 3), (1, 4), (0, 5)\}$ .
- (d) Now We have to allocate the kids between the three possible locations; there are 21 total outcomes in the sample space:

$$\begin{array}{ccccccc} (5, 0, 0) & (4, 1, 0) & (4, 0, 1) & (3, 2, 0) & (3, 1, 1) & (3, 0, 2) & (2, 3, 0) \\ (2, 2, 1) & (2, 1, 2) & (2, 0, 3) & (1, 4, 0) & (1, 3, 1) & (1, 2, 2) & (1, 1, 3) \\ (1, 0, 4) & (0, 5, 0) & (0, 4, 1) & (0, 3, 2) & (0, 2, 3) & (0, 1, 4) & (0, 0, 5) \end{array}$$

- (e) There could be from 0 to 5 kids in Hawkins. If there are  $k$  kids in Hawkins, there are  $\binom{5}{k}$  ways of selecting those kids. If we add up all the possibilities we get 32 outcomes:

$$\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 1 + 5 + 10 + 10 + 5 + 1$$

- (f) The amount of time is a continuous value, so it has the sample space  $S = [0, \infty)$ .
  - (g) There are 5 choices for the first arrival, 4 choices for the next arrival, and so on. That is, there are  $5 \times 4 \times 3 \times 2 = 120$  unique outcomes in the sample space.
  - (h) There are  $3 \cdot 2 = 6$  possible (body, mind) combinations for each kid, and there are 5 kids, so there are  $6^5 = 7776$  outcomes in the sample space.
2. (a) The probability that a kid's mind is not in their body is  $\frac{1}{2}$ , so the probability that no kids are in their right minds is  $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$ .
  - (b) It's easier to flip this around by computing the probability that none of the kids' (body, mind) locations match and taking the complement. In that case, there are 6 choices for the first kid, 5 choices for the second kid, down to 2 choices for the 5th kid. So the probability is

$$1 - \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{6 \cdot 6 \cdot 6 \cdot 6 \cdot 6} = 1 - \frac{720}{7776} = 1 - \frac{5}{54} = \frac{49}{54}$$

- (c) The probability that a given kid's body is in the Upside Down is  $\frac{1}{3}$ . We need to compute the probability that at least three kids are in the Upside Down, which is the sum of (exactly 3 kids) + (exactly 4 kids) + (exactly 5 kids). In the first case, there are  $\binom{5}{3} = 10$  ways of choosing three kids out of 5, and the probability of each outcome is  $\left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 = \frac{4}{243}$ . So the total probability is:

$$\binom{5}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2 + \binom{5}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 + \binom{5}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^0 = \frac{40}{243} + \frac{10}{243} + \frac{1}{243} = \frac{51}{243} = \frac{17}{81}$$

- (d) There are 3 physical locations and 2 mental locations. There are three ways the three girls could be in the same physical location, each with probability  $\frac{1}{27}$ , and two ways the two boys could be in the same mental location, each with probability  $\frac{1}{4}$ . So the total probability is  $3 \cdot \frac{1}{27} \cdot 2 \cdot \frac{1}{4} = \frac{1}{18}$ .

3. This is a little tricky. There are  $\binom{12}{5} = 792$  ways of choosing the 5 slots to put the kids in, but what fraction of these have no adjacencies? There are a few ways to approach the problem but one intuitive way is to build in the requirement that there is at least one empty slot between each filled slot. Suppose the order of the kids in slots is A, B, C, D, E. We can replace these by the symbols  $A\_B\_C\_D\_E$  where  $\_$  represents an empty slot. Together, these 5 symbols take up 9 of the original 12 slots, and we just need to insert 3 additional empty slots. So it's like choosing 3 places out of a shorter row of 8 slots to put the additional blanks, i.e.  $\binom{8}{3} = 56$  possibilities. So the probability of no adjacent filled slots is  $\frac{56}{792} = \frac{7}{99}$ .

There's also a way of solving this using dynamic programming. Note that it's not possible to put 5 kids in only 8 slots without adjacencies, and there's only one way to put 5 kids in 9 slots without adjacencies. What would happen if we added a 10th slot on one end of the 9 slots? Either it could be empty (in which case there is only one possibility for the remaining 9 slots), or it's filled, in which case the adjacent slot must be empty, and we need to fill the remaining 8 slots with 4 non-adjacent kids. This suggests a recurrence

$$Q(k, n) = Q(k, n-1) + Q(k-1, n-2)$$

where  $Q(k, n)$  is the number of non-adjacent possibilities to put  $k$  objects in  $n$  slots. You can build up a table to get to the same number as above ( $Q(5, 12) = 56$ ).